Physics 3221 Spring 2025: Practice Final

Grade:

1. Another Rocket Problem

Consider a rocket traveling in the x-direction in outer space (no gravity). The rocket starts with a mass m_0 and rest $v_0 = 0$ and expels fuel at a speed v_{ex} relative to the rocket. Set the origin as the position of the rocket at $t = 0$ (in other words, $x_0 = 0$).

(a) Find $v(t)$ for the rocket in terms of v_{ex} and m_0.

(b) Find $x(t)$ for the rocket in terms of v_{ex}, k and m_0, assuming $dm/dt = -k$.

(c) Find the time t_2 such that the rocket's mass is half its original mass, $m(t) = m_0/2$.

(d) What is the kinetic energy of the rocket at time t_2, once its mass is $m(t) = m_0/2$?

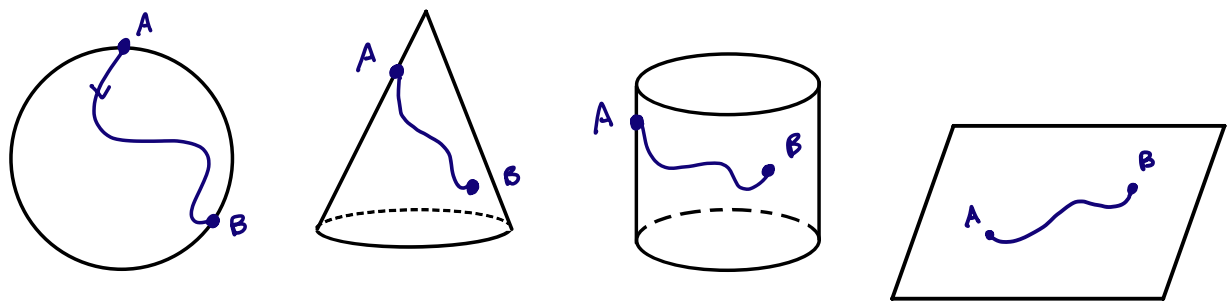
(e) How much work was done on (or done by) the rock between $t = 0$ and t_2? Just find the absolute value of the work, don't worry about the sign.

(f) How far did the rocket travel by time t_2?

(g) What is the average force acting on the rocket between $t = 0$ and $t = t_2$?

Grade:

2. **Geodesics** Calculate the geodesic (the shortest distance between two points) of a few surfaces: cone, sphere, cylinder, plane



(a) A sphere with radius R

(b) A circular cone with a surface defined by $\rho = \lambda z$

(c) A cylinder of radius R

(d) A flat plane

Figure 1: Several surfaces

(a) Find the shortest distance between point A at position $(R, 0, 0)$ and B at position (R, θ_B, ϕ_B) following a path on the surface of a sphere shown in Fig. 1a.

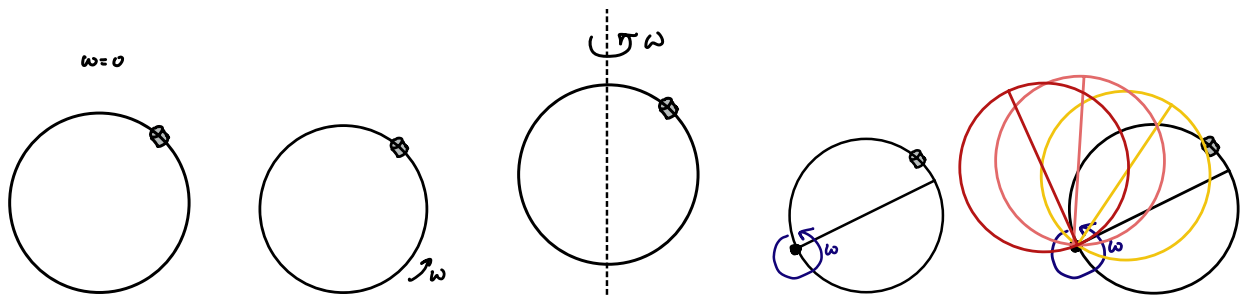
(b) Find the shortest distance between point A at position (ρ_A, π, z_A) and B at position (ρ_B, ϕ_B, z_B) following a path on the surface of a cone shown in Fig. 1b. Once you get to trying to determine the two unknown coefficients by plugging in ϕ_b, z_b, z_A, you can stop, as the expressions get complicated.

(c) Find the shortest distance between point A at position (R, π, z_A) and B at position (R, ϕ_B, z_B) following a path on the surface of the cylinder shown in Fig. 1c.

(d) Find the shortest distance between point A at position $(0, 0, 0)$ and B at position (x_B, y_B, z_B) following a path on the surface of a cone shown in Fig. 1d.

3. The Rotating Hoop

Consider a bead of mass m on a hoop of radius R, where the hoop can rotate in many different ways, see Fig. 2. The hoop's angular velocity is given by ω, which is not necessarily the speed of the bead. In all cases, the hoop experiences gravity pointing downwards (not into or our of the page).



(a) (b) (c) (d)

Figure 2: A bead of mass m on a hoop rotating many different ways. In all figures, gravity is pointing downwards (not into or out of the page). In Fig. 2a, the hoop is not rotating ($\omega = 0$). In Fig. 2b, the hoop is rotating about an axis pointing out of the paper and running through the center of the hoop. In Fig 2c, the figure is rotating about the vertical axis. In Fig. 2d, the hoop is rotating about a point on its circumference, around an axis pointing out of the page. The general setup is shown on the left of Fig. 2d, while this rotation is demonstrated on by showing snapshots as the rotation progresses on the right.

(a) Write down the Lagrangian for all four cases

(b) Write down the equations of motion for all four cases

(c) For cases shown in Fig 2a, Fig 2b, and Fig 2c, find the equilibrium points of the system, and identify whether they are stable or unstable. If they are stable, find the frequency of small oscillations about the equilibrium point.

Grade:

4. The Pulley with a Spring

Consider the pulley system of three masses shown in Fig. 3. The masses m_1 and m_2 are connected by a rope of fixed length, which goes over a massless pulley. The mass m_3 is connected to m_2 by a spring. The spring has a spring constant k and an equilibrium length l_0. For simplicity, assume all three masses have negligible height and width compared to the length of the rope.

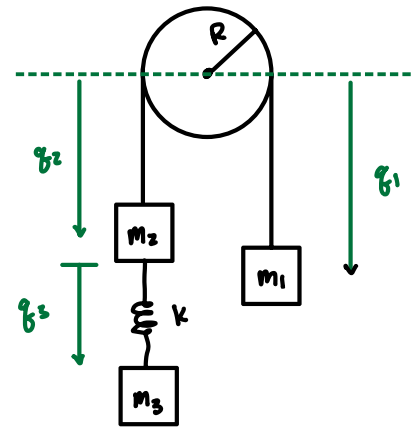


Figure 3: A pulley system of three masses. The masses m_1 and m_2 are connected by a rope, while m_3 hangs from a spring of spring constant k, equilibrium length l_0.

(a) Write down the Lagrangian in terms of the minimal number of generalized coordinates

(b) Use the Euler Lagrange equations to find the equation of motion of the system.

(c) Simplify the equations of motion. If they are coupled, try to uncouple them (in other words, find an expression for $\ddot{q}_3$ without q_1-dependence and vice versa).

(d) Next, use the method of Lagrange multipliers and modified Lagrange equations to explain how you would find the tension in the rope. You don't have to solve the full set of differential equations, but explain the steps.

Grade:

5. The Driven Damped Oscillator

Consider a mass m attached to a horizontal spring with spring constant k, only allowed to move in the horizontal direction. The mass is subject to a linear drag force $\vec{F} = -b\vec{v}$, where b is the drag coefficient, and the mass is subject to a driving force going in the horizontal direction. The driving force is given by:

$$F(t) = F_0 t^2, \qquad t \in (-\tau/2, \tau/2) \tag{1}$$

$$F(t+\tau) = F(t) \tag{2}$$

(a) Write down the equation of motion for the mass given both the drag and driving forces.

(b) Now let's turn the driving force off. Find the solution to the equation of motion for all cases: underdamped, overdamped, and critically damped. Let $x_0 = 0$ and leave v_0 undetermined. What is the amplitude and phase in terms of m, b, k and v_0?

(c) Sketch the shape of $F(t)$

(d) Find the Fourier expansion of $F(t)$

(e) Find the general solution for $x(t)$ given the driving force shown in Eq. (2). Provide the solution for all cases: underdamped, overdamped, and critically damped.

Formulas (just for the Practice Final)

Newton's 2nd Law

$$\vec{F} = m\ddot{\vec{r}}$$

$$F_x = m\ddot{x} \qquad F_y = m\ddot{y} \qquad F_z = m\ddot{z}$$

$$F_r = m(\ddot{r} - r\dot{\phi}^2) \qquad F_\phi = m(r\ddot{\phi} + 2\dot{r}\dot{\phi})$$

$$F_\rho = m(\ddot{\rho} - \rho\dot{\phi}^2) \qquad F_\phi = m(\rho\ddot{\phi} + 2\dot{\rho}\dot{\phi}) \qquad F_z = m\ddot{z}$$

Coordinate Systems

Cartesian: $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$

$d\vec{r} = dx\hat{x} + dy\hat{y} + dz\hat{z}$

Cylindrical : $\vec{r} = \rho\hat{\rho} + z\hat{z}$

$d\vec{r} = d\rho\hat{\rho} + \rho d\phi\hat{\phi} + dz\hat{z}$

Spherical: $\vec{r} = r\hat{r}$

$d\vec{r} = dr\hat{r} + rd\theta\,\hat{\theta} + r\sin\theta\, d\phi\,\hat{\phi}$

Fourier Series

$$f(t) = \sum_{n=0}^{\infty} [a_n \cos(n\omega t) + b_n \sin(n\omega t)]$$

$$a_0 = \frac{1}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\,,$$

$$a_n = \frac{2}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\cos(n\omega t)dt\,, \qquad n > 0$$

$$b_n = \frac{2}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\sin(n\omega t)dt\,, \qquad n > 0$$

$$e^{i\theta} = \cos\theta + i\sin\theta \qquad \text{(Euler's formula)}$$

Hook's Law & Simple Harmonic Motion

$$F = -kx \Leftrightarrow U = \frac{1}{2}kx^2$$

$$x(t) = A\cos(\omega t - \delta)$$

Driven Oscillator

$$A^2 = \frac{f_0^2}{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2)}$$

$$\delta = \tan^{-1}\left(\frac{2\beta\omega}{\omega_0^2 - \omega^2}\right)$$

Work

$$W = \int_{\vec{r}_0}^{\vec{r}} \vec{F}(\vec{r}\,') \cdot d\vec{r}\,'$$

Conservative forces

$$U(\vec{r}) = -\int_{\vec{r}_0}^{\vec{r}} \vec{F}(\vec{r}\,') \cdot d\vec{r}\,'$$

$$F = -\vec{\nabla}U$$

$$\vec{\nabla} \times \vec{F} = 0$$

Useful Integrals

$$\int x\sin(ax)dx = -\frac{x}{a}\cos(ax) + \frac{1}{a^2}\sin(ax)$$

$$\int x\cos(ax)dx = \frac{x}{a}\sin(ax) + \frac{1}{a^2}\cos(ax)$$

Gradient and Curl

$$\vec{\nabla}f = \hat{x}\frac{\partial f}{\partial x} + \hat{y}\frac{\partial f}{\partial y} + \hat{z}\frac{\partial f}{\partial z}$$

$$\vec{\nabla}f = \hat{r}\frac{\partial f}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial f}{\partial \theta} + \hat{\phi}\frac{1}{r\sin\theta}\frac{\partial f}{\partial \phi}$$

$$\vec{\nabla}f = \hat{\rho}\frac{\partial f}{\partial r\rho} + \hat{\theta}\frac{1}{\rho}\frac{\partial f}{\partial \theta} + \hat{z}\frac{\partial f}{\partial z}$$

$$\vec{\nabla} \times \vec{A} = \hat{x}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \hat{y}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \hat{z}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$$

Euler-Lagrange Equation

$$S = \int_{x_1}^{x_2} f[y(x), y'(x), x]dx$$

$$\Rightarrow \frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y'} = 0$$

Lagrange Equation

$$\mathcal{L} = T - U \quad T = \frac{1}{2}mv^2 = \frac{1}{2}m\vec{v}\cdot\vec{v}$$

$$S = \int_{x_1}^{x_2} \mathcal{L}(q_1, .., q_i, \dot{q}_1, ..., \dot{q}_i, t)dt$$

$$\Rightarrow \frac{\partial \mathcal{L}}{\partial q_i} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i}$$

Lagrange Multiplier

$$f(x_1, x_2) = const.$$

$$\frac{\partial \mathcal{L}}{\partial x_i} + \lambda\frac{\partial f}{\partial x_i} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}_i}$$

$$F_i = \lambda\frac{\partial f}{\partial x_i}$$